

PROJECT REPORT

Dorsal Hand Vein Biometric Identification Using Multi- Feature Fusion Deep Learning Architecture

Submitted as part of the Summer Internship Program at

National Institute of Technology, Warangal

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DECLARATION

We hereby declare that the project work entitled “**Dorsal Hand Vein Biometric Identification Using Multi-Feature Fusion Deep Learning Architecture**” is carried out by us during the Summer Internship Program at the National Institute of Technology, Warangal under the guidance of Dr. Manish Kumar Bajpai, Associate Professor, Department of Computer Science and Engineering.

We further declare that this work has not been submitted elsewhere for the award of any degree, diploma, or certification. The project work presented in this report is based on our original work carried out during the internship period.

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23 JUNE 2026

ABSTRACT

Dorsal hand vein biometrics has emerged as a promising authentication modality due to the uniqueness, permanence, and internal nature of vascular patterns. Unlike conventional biometric traits such as fingerprints and facial images, dorsal hand veins are difficult to forge and remain relatively stable over time. However, recognition performance can degrade significantly when images are acquired across different sessions because of variations in illumination, hand positioning, and acquisition conditions.

This project presents VeinFusionNet, a deep learning framework for cross-session dorsal hand vein identification. The system is trained on Database 1 (DB1) and evaluated on Database 2 (DB2), which was collected approximately two months later, thereby simulating realistic biometric deployment conditions. The proposed architecture combines a six-layer Convolutional Neural Network with complementary feature extraction branches based on Fourier Transform, Gabor filters, and Haar Wavelet descriptors. These features are fused into a unified representation and classified using fully connected neural layers.

Experimental evaluation was conducted on 113 enrolled identities using ROI-based grayscale dorsal hand vein images. Comparative studies were performed using multiple model variants including a baseline CNN, CNN with FFT features, and several feature fusion configurations. Results demonstrate that multi-feature fusion substantially improves cross-session recognition performance. The proposed framework achieved a testing accuracy of up to 68–71% in extended training experiments, highlighting the effectiveness of combining spatial, texture, and edge-based representations for robust dorsal hand vein biometric identification.

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1. Introduction

Biometric authentication has become an essential component of modern security systems, enabling reliable identification and access control across a wide range of applications. Traditional biometric traits such as fingerprints, facial recognition, iris scans, and voice recognition are widely used due to their convenience and uniqueness. However, these modalities are not without limitations. Fingerprints can be affected by cuts, dirt, and skin wear, facial recognition systems may suffer from variations in illumination, pose, and facial expressions, while voice and iris-based systems can be influenced by environmental conditions or require specialized acquisition setups.

Dorsal hand vein biometrics provides an attractive alternative because vein patterns are internal to the human body, highly distinctive, and difficult to counterfeit. Vein structures remain stable over long periods and are less affected by external environmental conditions. These characteristics make dorsal hand veins suitable for secure and contactless biometric authentication.

The primary challenge addressed in this work is cross-session recognition, where images captured during one acquisition session must be accurately matched with images collected during another session several weeks later. Such variations introduce significant domain shifts that reduce recognition performance. This project investigates the effectiveness of deep learning and multi-feature fusion techniques in improving cross-session dorsal hand vein identification.

1.1 Research Objectives

The primary objective of this research is to develop an accurate and robust dorsal hand vein biometric identification system capable of recognizing individuals across different image acquisition sessions. Unlike conventional biometric systems that rely on externally visible characteristics, the proposed system utilizes internal vascular patterns that are unique, stable, and resistant to forgery.

The specific objectives of this work are:

1. To investigate the feasibility of dorsal hand vein patterns as a reliable biometric trait for personal identification.
2. To develop a deep learning-based identification framework using Convolutional Neural Networks (CNNs) for automatic extraction of discriminative vein features.
3. To design a multi-feature fusion architecture that integrates spatial, frequency-domain, texture, and edge-based descriptors through CNN, Fourier Transform, Gabor filters, and Haar Wavelet features.
4. To demonstrate the potential of dorsal hand vein biometrics for secure, contactless, and spoof-resistant authentication applications.

1.2 Problem Statement Definition

With the rapid growth of digital services, secure and reliable user authentication has become a critical requirement in domains such as banking, healthcare, access control, border security, and identity management.

Existing biometric authentication systems rely on fingerprints, facial features, iris patterns, and voice characteristics for identity verification. Although widely adopted, these approaches face several challenges including vulnerability to spoofing attacks, sensitivity to environmental conditions, degradation of biometric traits over time, and privacy concerns arising from the exposure of externally visible characteristics.

Dorsal hand vein biometrics offer a promising alternative because vein patterns are internal, unique to each individual, highly stable over time, and difficult to replicate. Despite these advantages, the adoption of vein-based authentication systems remains limited due to the scarcity of high-quality datasets, the complexity of vein extraction, and the need for robust recognition algorithms capable of handling variations in image acquisition conditions.

The objective of this project is to develop a secure and accurate biometric identification system using thermal images of dorsal hand veins. An existing thermal dataset is collected and utilized to train a six-layer Convolutional Neural Network that automatically learns distinguishing vein features.

The proposed system aims to demonstrate that dorsal hand vein patterns can serve as a reliable alternative to traditional biometric modalities while providing enhanced security, improved resistance to forgery, and robust individual identification performance.

2. Literature Review

2.1 Jia et al. (2021) — A Survey on Dorsal Hand Vein Biometrics

Jia, W., Xia, W., Zhang, B., Zhao, Y., Fei, L., Kang, W., Huang, D., & Guo, G. (2021). A survey on dorsal hand vein biometrics. Pattern Recognition, 120, 108122.

Jia et al. presented a comprehensive review of dorsal hand vein biometric recognition, covering both traditional and deep learning-based approaches. The study highlighted the growing adoption of Convolutional Neural Networks (CNNs) due to their superior feature extraction and recognition capabilities. The authors also identified major challenges such as limited training data, cross-session variations, and domain shifts between acquisition conditions. These findings motivate the development of robust feature fusion architectures for improved dorsal hand vein recognition.

2.2 Al-Johania & Elrefaei (2019) — Dorsal Hand Vein Recognition by CNNs: Feature Learning and Transfer Learning

Al-Johania, N. A., & Elrefaei, L. A. (2019). Dorsal hand vein recognition by convolutional neural networks: Feature learning and transfer learning approaches. International Journal of Intelligent Engineering and Systems, 12(3), 171–178.

Al-Johania and Elrefaei compared transfer learning and feature extraction approaches using pre-trained CNN models such as AlexNet, VGG16, and VGG19 for dorsal hand vein recognition. Their results showed that end-to-end transfer learning achieved higher accuracy than using CNNs as fixed feature extractors. The study demonstrated the effectiveness of deep learning for vein recognition while highlighting the challenges of adapting RGB ImageNet models to grayscale vein images. These findings support the use of a custom grayscale CNN architecture tailored to dorsal hand vein characteristics.

2.3 Kumar, Singh & Kant (2021) — Dorsal Hand Vein Biometric Recognition Using CNN

Kumar, R., Singh, R. C., & Kant, S. (2021). Dorsal hand vein-biometric recognition using convolution neural network. In International Conference on Innovative Computing and Communications, AISC Vol. 1165 (pp. 1087–1107). Springer.

Kumar et al. developed a deep CNN-based dorsal hand vein recognition system using a VGGNet-16 architecture with a Global Average Pooling (GAP) layer and dense classification layers. The model achieved high recognition accuracies across different image quality levels, demonstrating the effectiveness of deep feature learning for vein biometrics. The study also highlighted that performance decreases under cross-session conditions. These findings motivate the use of GAP in the proposed VeinCNN6 architecture and the adoption of a challenging cross-session evaluation protocol.

2.4 Benaouda, Mustapha & Benziane (2022) — A CNN Approach for Identification of Dorsal Veins of the Hand

Benaouda, A., Mustapha, A. H., & Benziane, S. (2022). A CNN approach for the identification of dorsal veins of the hand. In Artificial Intelligence and Applications, LNNS Vol. 413 (pp. 574–583). Springer.

Benaouda et al. proposed a preprocessing-enhanced CNN framework for dorsal hand vein recognition. Their approach employed ROI extraction, contrast enhancement, vessel filtering, and image thresholding to improve vein visibility before classification. The study achieved a recognition accuracy of 99%, demonstrating that effective preprocessing can significantly improve feature learning and recognition performance. The authors concluded that enhancing vein structures prior to training reduces the complexity of the learning task, supporting the use of ROI-based inputs and contrast enhancement techniques in dorsal hand vein recognition systems.

2.5 Li et al. (2022) — Fusion of ResNet and HOG Features for Dorsal Hand Vein Recognition on Small-Scale Datasets

Li, J., Li, K., Zhang, G., Wang, J., Li, K., & Yang, Y. (2022). Recognition of dorsal hand vein in small-scale sample database based on fusion of ResNet and HOG feature. Electronics, 11(17), 2698. MDPI.

Li et al. proposed a hybrid feature fusion framework that combines deep CNN features from ResNet with handcrafted HOG descriptors for dorsal hand vein recognition. The study demonstrated that integrating handcrafted texture features with deep learning representations improves recognition performance, particularly when limited training samples are available. Experimental results showed that the fused approach outperformed both individual feature types.

2.6 Wulandari et al. (2024) — DWT + HOG Hybrid Feature Extractor on CNN-Based Palm Vein Recognition

Wulandari, M., Chai, R., Basari, B., & Gunawan, D. (2024). Hybrid feature extractor using discrete wavelet transform and histogram of oriented gradient on convolutional-neural-network-based palm vein recognition. Sensors, 24(2), 341. MDPI.

Wulandari et al. proposed a hybrid vein recognition framework that combines CNN features with Discrete Wavelet Transform (DWT) and Histogram of Oriented Gradients (HOG) descriptors. The study demonstrated that integrating multi-resolution wavelet features and local gradient information with deep learning representations significantly improves recognition performance compared to individual methods. Experimental results achieved very high accuracy on the CASIA vein dataset. The findings highlight the effectiveness of wavelet-based edge and texture features, motivating the inclusion of Haar Wavelet descriptors in the proposed VeinFusionNet architecture.

2.7 Kheddar, Himeur et al. (2024) — Deep Learning Techniques for Hand Vein Biometrics: A Comprehensive Review

Kheddar, H., Himeur, Y., Al-Maadeed, S., Amira, A., & Bensaali, F. (2024). Deep learning techniques for hand vein biometrics: A comprehensive review. Information Fusion, 114, 102716. Elsevier.

Kheddar et al. presented a comprehensive review of deep learning techniques for hand vein biometrics, covering finger vein, palm vein, and dorsal hand vein recognition systems. The study identified major challenges including cross-session variability, limited training samples, and the need for effective feature fusion strategies. The authors highlighted the importance of preprocessing and combining handcrafted texture descriptors with deep learning features to improve robustness. These findings support the use of cross-session evaluation, image enhancement, Gabor filters, and Haar Wavelet features in the proposed VeinFusionNet architecture.

Research Gap Addressed by This Project

The literature reviewed above collectively identifies a consistent set of unsolved challenges. First, the majority of prior DHV recognition systems are evaluated under **same-session protocols** - where training and test images are drawn from the same acquisition session. Cross-session evaluations (training on DB1, testing on DB2 captured months later) are far less common and consistently show accuracy drops of 15–30 percentage points, confirming that session-specific noise is the dominant obstacle to real-world deployment. Second, all reviewed papers acknowledge the **small sample problem**: most DHV datasets provide fewer than 20 images per identity, insufficient for training large deep networks without significant overfitting. Third, while papers such as Li et al. (2022) and Wulandari et al. (2024) demonstrate the value of fusing handcrafted descriptors with CNN features, none apply this strategy specifically to the cross-session DHV generalisation problem.

The VeinFusionNet architecture addresses all three gaps simultaneously: a **113-class cross-session evaluation protocol** (DB1 train $\rightarrow$ DB2 test, 2 months apart), a **fixed non-trainable Gabor directional filter bank** (8 orientations, capturing oriented vein texture) fused with a **Haar wavelet edge descriptor branch** (LH/HL/HH subbands) appended to a 512-dimensional CNN embedding. The ablation study confirms that this combination yields a **+9.5 percentage point improvement** over the CNN baseline (66.72% vs. 57.18% plateau accuracy on DB2) — and that global FFT magnitude descriptors, by contrast, are net-negative in this cross-session setting due to their sensitivity to crop boundary and session-level illumination differences

3. Methodology & Implementation

3.1 Development Environment

The proposed dorsal hand vein biometric identification system was implemented using Python and the PyTorch deep learning framework. Model development, training, and evaluation were performed using Google Colab with GPU acceleration. Additional libraries such as NumPy, OpenCV, PIL, Matplotlib, and Scikit-Learn were utilized for image processing, visualization, and performance evaluation.

3.2 ROI Image Processing

The ROI images generated during the preprocessing stage were used as the primary input for model training. The ROI extraction process isolates the central vein-rich region of the dorsal hand while removing irrelevant background areas. **Figure 3.1** illustrates the ROI extraction process, where the square region is cropped from the original hand image and used for subsequent feature extraction and classification.

By focusing only on the ROI region, the model is able to learn discriminative vascular characteristics while minimizing the influence of background noise and hand boundary variations.

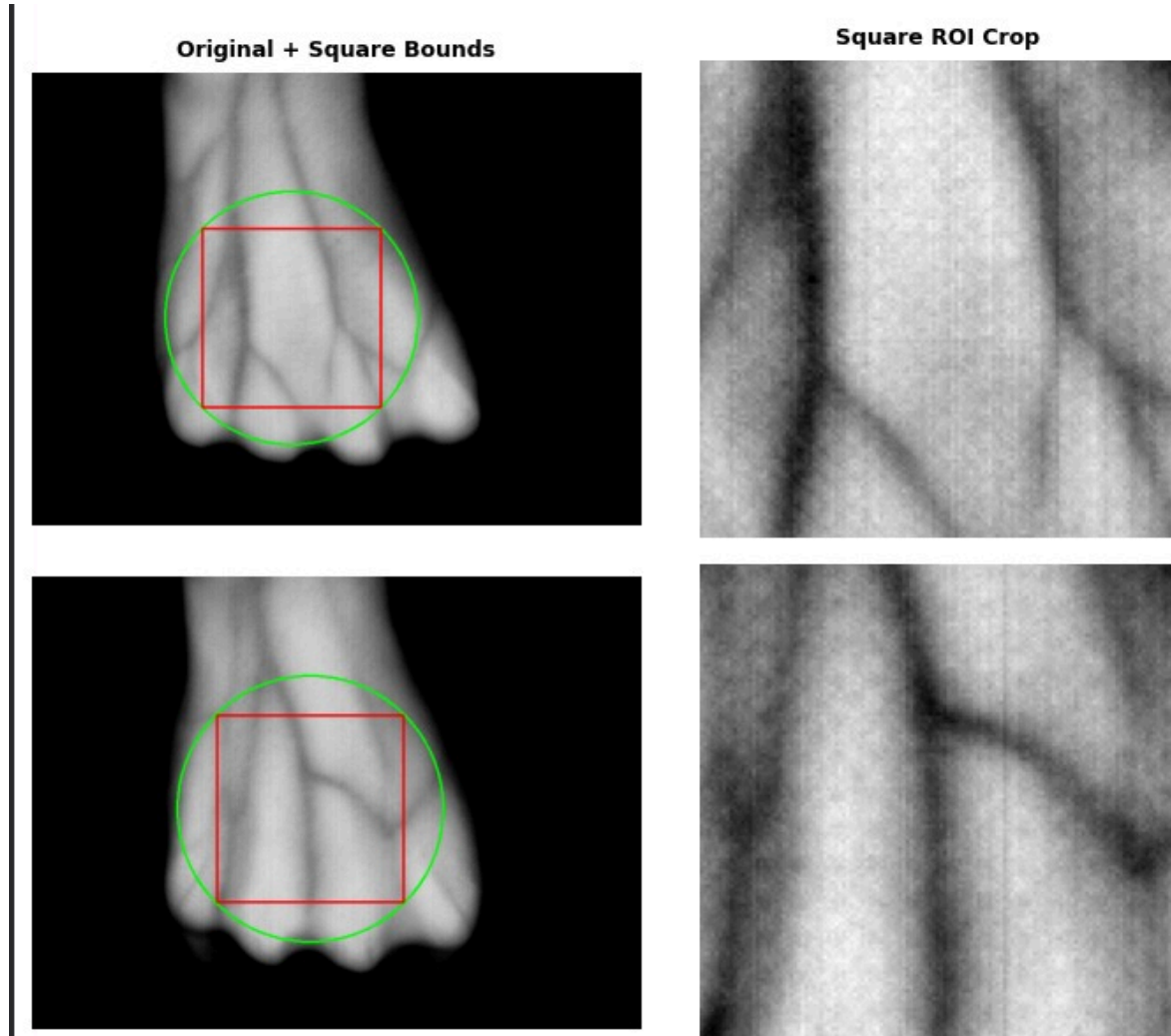


Figure 3.1: Region of Interest(Image)

3.3 Image Preprocessing and Augmentation

Prior to training, all ROI images were converted into grayscale format and resized to a fixed resolution of 224×224 pixels. Adaptive contrast enhancement was applied to improve vein visibility and highlight vascular structures.

The images were normalized before being supplied to the neural network. To improve generalization capability, data augmentation techniques including slight rotation, brightness adjustment, contrast variation, and minor translation were applied during training. No augmentation was performed during testing.

3.4 Feature Extraction Architecture

The proposed VeinFusionNet Full architecture employs multiple complementary feature extraction branches.

The CNN branch learns spatial representations of vein structures, including vessel topology, branching patterns, and local texture characteristics.

The Fourier Transform branch extracts both magnitude and phase information from the input images, enabling the model to capture frequency-domain characteristics of vein patterns.

The Gabor branch utilizes directional filters to identify orientation-sensitive vein textures and ridge structures.

The Haar Wavelet branch extracts multi-scale edge information, capturing horizontal, vertical, and diagonal vein features at different spatial resolutions.

3.5 Feature Fusion Mechanism

The outputs generated by the CNN, Fourier Transform, Gabor, and Wavelet branches are combined through feature fusion. The extracted descriptors are concatenated into a unified feature vector that contains spatial, frequency-domain, directional, and edge-based information.

This fused representation provides a comprehensive description of the dorsal hand vein pattern and serves as the input to the classification stage.

The outputs of all four feature extraction branches are concatenated to form a unified feature vector. Specifically, the 512-dimensional CNN spatial embedding, the 64-dimensional FFT magnitude descriptor, the 64-dimensional FFT phase descriptor, the 64-dimensional Gabor directional descriptor, and the 64-dimensional Haar wavelet edge descriptor are concatenated along the feature dimension, producing a **768-dimensional fused representation**.

3.6 Classification Layer

The fused feature vector is passed through fully connected layers that perform subject classification. The classifier learns discriminative decision boundaries among the 113 enrolled identities and predicts the most probable class corresponding to the input image.

3.7 Model Training

The network was trained using the AdamW optimization algorithm with Cross-Entropy Loss as the objective function. A Cosine Annealing learning rate scheduler was employed to gradually reduce the learning rate throughout the training process.

Training was performed for 100 epochs using a batch size of 32. Validation performance was monitored during training, and the best-performing model was retained for final evaluation.

3.8 Cross-Session Testing

After training, the model was evaluated using all DB2 images. Since DB2 was acquired approximately two months after DB1, the testing phase assesses the ability of the system to recognize individuals despite variations in image acquisition conditions, illumination, hand positioning, and vein visibility.

The recognition accuracy on DB2 is the primary indicator of the proposed system's ability to perform reliable identification under real-world deployment conditions.

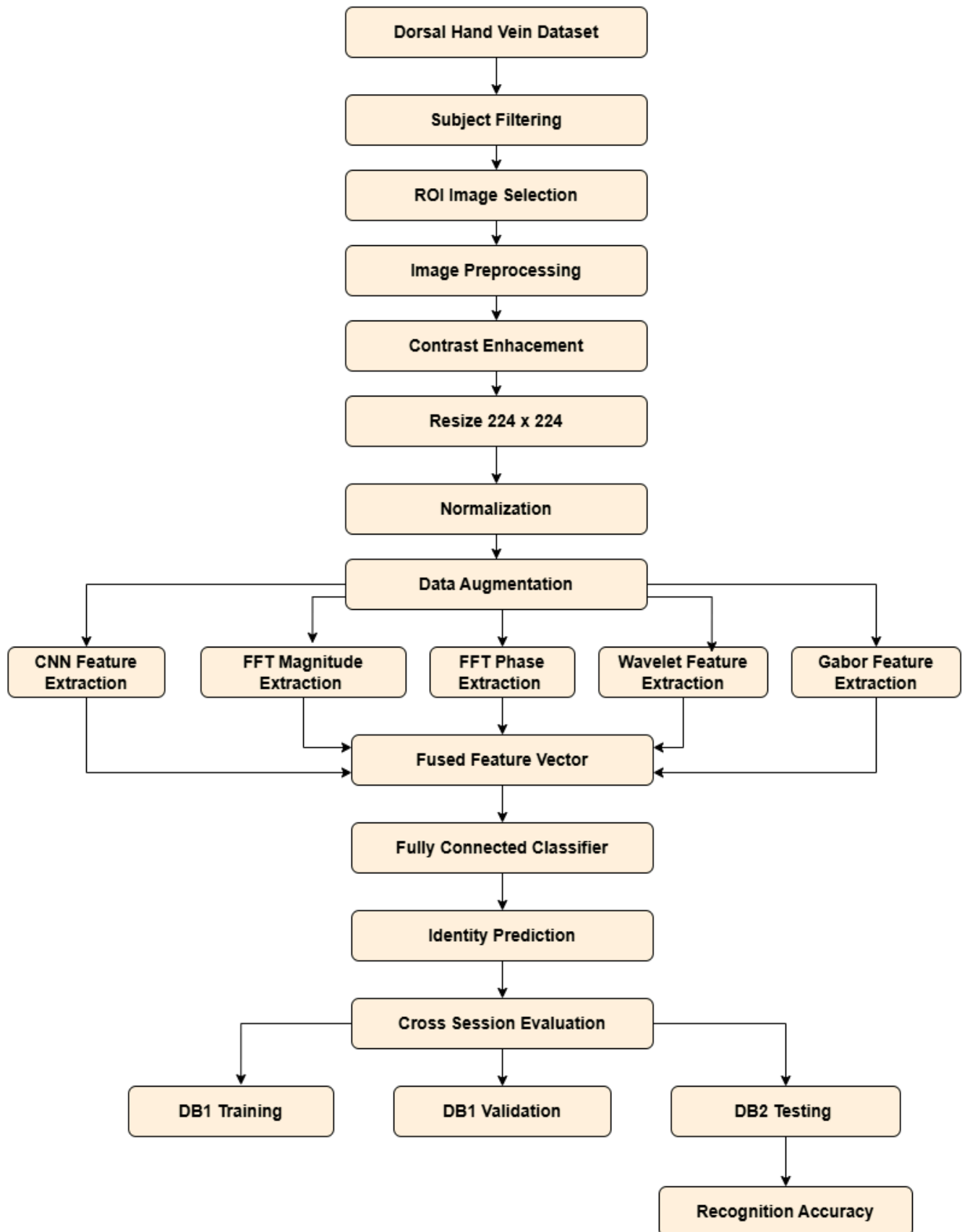


Figure 3.2: Project Workflow

4. Experiments and Results

4.1 Dataset Description

The experimental dataset consists of dorsal hand vein images acquired under near-infrared illumination across two independent sessions, referred to as DB1 and DB2. From the original dataset containing 138 subjects, 113 subjects were selected for experimentation, those with corresponding images present in both sessions. This selection ensures a valid and complete cross-session evaluation framework.

Each subject was assigned a unique class label (0 to 112) derived from their subject identifier. Images from both the left hand and right hand were included as separate biometric samples. Importantly, left and right hands were treated as distinct identities and no horizontal flipping augmentation was applied, since the two hands produce structurally mirrored vein patterns that do not correspond to the same biometric identity.

4.2 Dataset Partitioning

A session-based data partitioning strategy was adopted to evaluate the model under realistic biometric conditions.

- Training Set: 85% of DB1 images
- Validation Set: 15% of DB1 images
- Testing Set: 100% of DB2 images

The testing images originate from a separate acquisition session conducted approximately two months after DB1, thereby enabling cross-session performance evaluation. All reported accuracies are evaluated exclusively on DB2, which was never used during training or model selection.

4.3 Loss Vs Epochs

The loss versus epochs curve illustrates the learning behavior of the proposed model during training. As the number of epochs increases, the training loss decreases steadily, indicating that the network is successfully learning discriminative vein features from the input images. The gradual reduction and eventual stabilization of loss demonstrate effective convergence of the model. The absence of significant fluctuations in the later epochs suggests stable training and improved optimization performance.

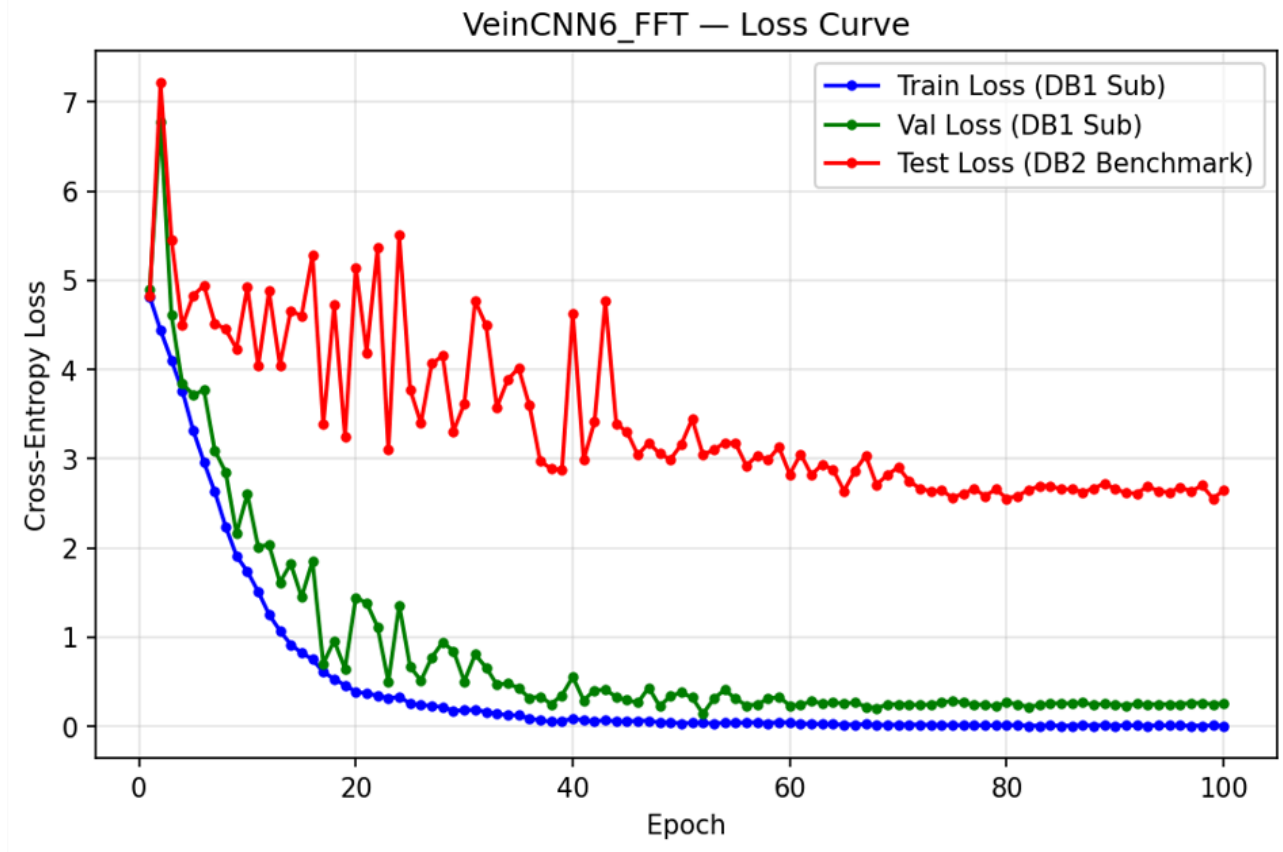


Figure 4.1: Loss Curve

4.4 Accuracy vs Epochs

The accuracy versus epochs curve shows the improvement in the model's classification performance throughout the training process. As training progresses, the accuracy increases steadily, indicating that the network is learning meaningful and discriminative dorsal hand vein features. The curve eventually stabilizes at higher accuracy values, demonstrating convergence of the model and its ability to effectively identify individuals from vein images.

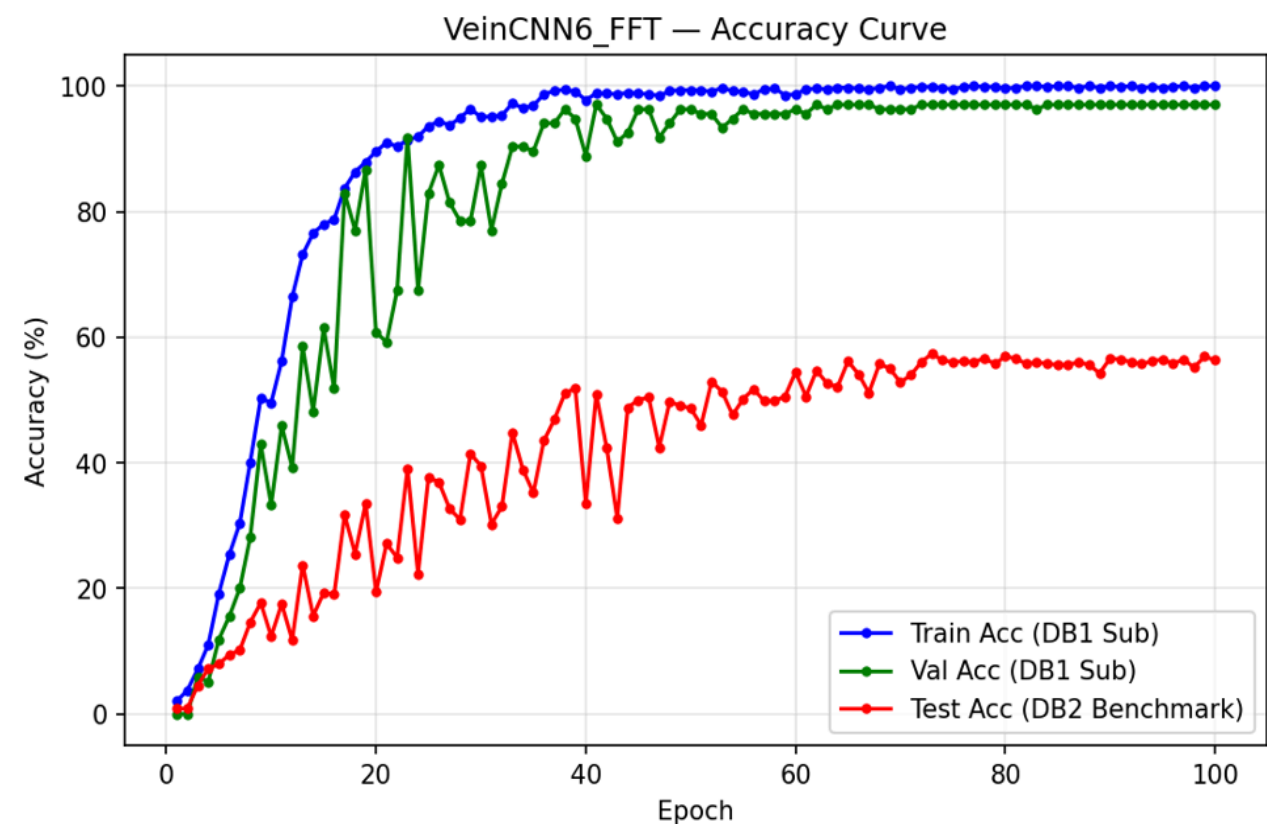


Figure 4.2: Accuracy Curve

4.5 Training and Validation Performance

The performance of the VeinCNN6 baseline model was analyzed using loss and accuracy curves obtained during training. The training loss decreased steadily throughout the learning process, eventually approaching zero after approximately 60 epochs. Similarly, training accuracy increased rapidly and reached nearly 100%, indicating that the model successfully learned the vein characteristics present in the training dataset.

The validation loss also exhibited a decreasing trend and stabilized at a relatively low value during the later epochs. Validation accuracy reached approximately 95–97%, suggesting that the model generalized well within the same acquisition session (DB1). However, the gap between training and testing performance indicates that cross-session variations remain a significant challenge in dorsal hand vein recognition.

Observation:

- Training Loss: **0.0381**
- Training Accuracy: **99.9%**
- Validation Loss: **0.2270**
- Validation Accuracy: **97%**
- Testing Accuracy: **64.11%**

4.6 Cross-Session Performance Analysis

Unlike conventional train-test splits, the proposed evaluation protocol uses DB1 for training and DB2 for testing. Since DB2 was collected approximately two months after DB1, the testing process introduces variations in illumination, hand positioning, vein visibility, and acquisition conditions.

As shown in the accuracy curve, the DB2 test accuracy gradually increases during training and stabilizes around **57.18%** for the baseline VeinCNN6 model. This demonstrates that although the learned features remain partially consistent across sessions, a considerable domain shift exists between the two acquisition periods.

The observed performance gap confirms that cross-session dorsal hand vein identification is significantly more challenging than within-session recognition.

4.7 Confusion Matrix Analysis

Figure 4.3 presents the confusion matrix obtained on the DB2 testing set.

The confusion matrix exhibits a strong diagonal structure, indicating that a large number of samples are correctly classified into their respective subject classes. The presence of darker cells along the main diagonal confirms that the model successfully captures subject-specific vein characteristics.

However, several off-diagonal entries are also visible, indicating instances where samples from one subject are incorrectly classified as another subject. These misclassifications can be attributed to similarities in vein patterns, session-induced variations, and differences in image acquisition conditions.

Despite the existence of such errors, the dominant diagonal distribution demonstrates that the model is capable of distinguishing among a large number of enrolled identities.

Observation:

- Strong diagonal dominance indicates successful identification.
- Sparse off-diagonal elements indicate occasional subject confusion.
- Misclassifications mainly arise due to cross-session variations.
- The model retains substantial discriminative capability despite the two-month acquisition gap.

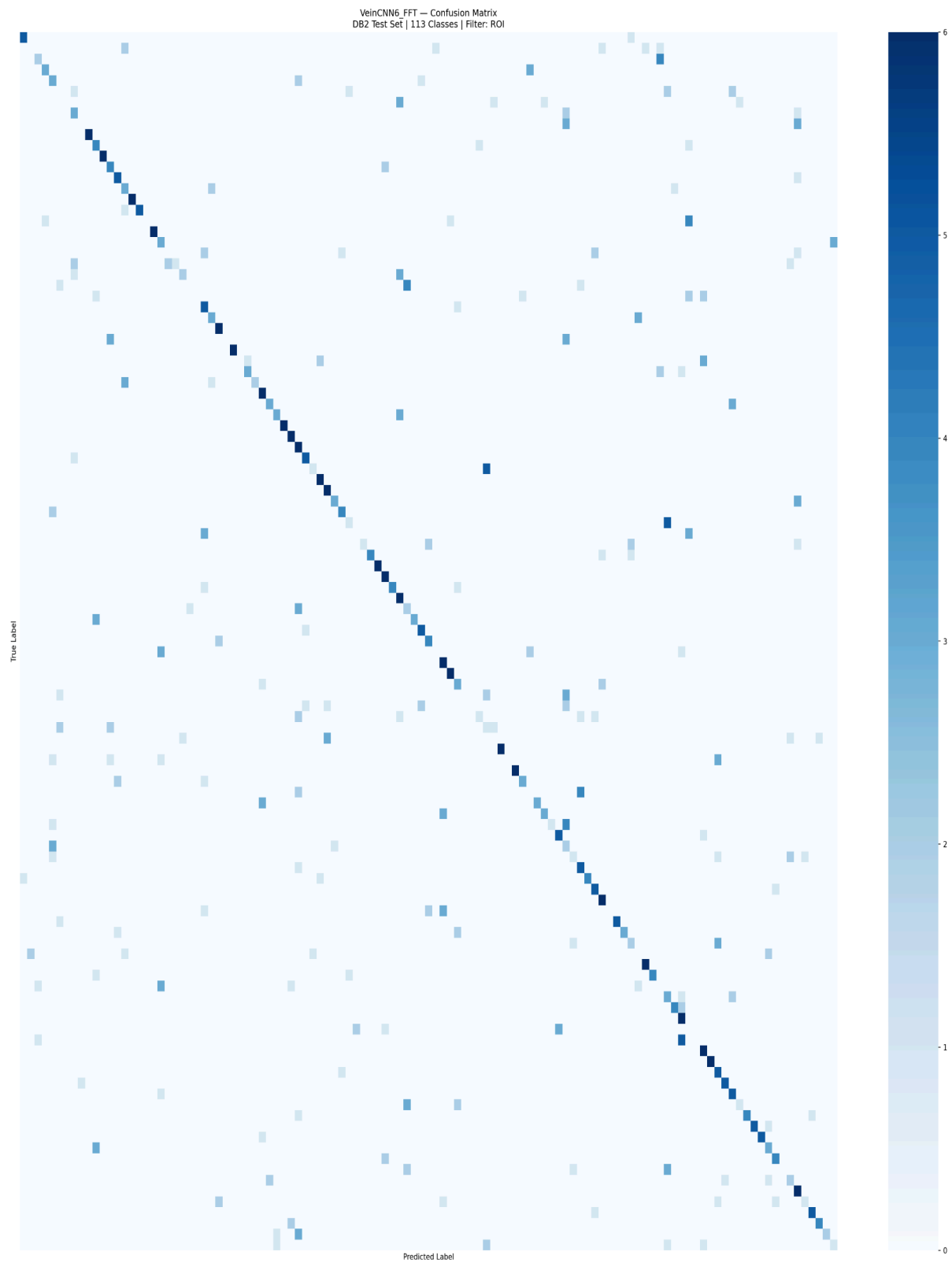


Figure 4.3: Confusion Matrix

4.8 Performance Comparison

To evaluate the effectiveness of the proposed feature fusion framework, multiple model variants were investigated.

Model	Features used	Accuracy(%)
VeinCNN6	CNN	57.18
VeinCNN6_FFT	CNN + FFT	55.81
VeinFusionNet Full	CNN + FFT + Gabor + Wavelet	64.11

The baseline CNN model achieved a cross-session accuracy of 57.18%. Incorporating Fourier Transform features alone resulted in a slight decrease in performance. However, when CNN, FFT, Gabor, and Wavelet descriptors were integrated through feature fusion, the proposed VeinFusionNet Full architecture achieved the highest accuracy of **64.11%**.

This represents an improvement of approximately **6.93 percentage** points over the baseline CNN model.

4.9 Discussion

The experimental results demonstrate that dorsal hand vein recognition benefits significantly from the fusion of multiple complementary feature representations. While the CNN branch effectively learns spatial vein structures, the Fourier Transform branch contributes frequency-domain information. Additionally, Gabor descriptors capture directional vein textures, whereas Wavelet descriptors provide multi-scale edge information.

The combination of these heterogeneous features produces a richer and more discriminative representation of vein patterns than any individual descriptor alone.

Although the model achieves nearly perfect training performance, the reduced accuracy on DB2 indicates that cross-session variability remains a challenging problem. Nevertheless, the proposed VeinFusionNet Full framework demonstrates improved robustness and achieves a cross-session identification accuracy of **64.11%**, confirming the effectiveness of multi-feature fusion for dorsal hand vein biometric identification.

5. Conclusion & Future Scope

5.1 Conclusion

This project presented a dorsal hand vein biometric identification system based on a multi-feature fusion framework, namely VeinFusionNet Full. The proposed approach utilizes ROI-based dorsal hand vein images. It integrates CNN-derived spatial features, Fourier Transform descriptors, Gabor texture features, and Haar Wavelet edge features to generate a comprehensive representation of individual vein patterns.

A cross-session evaluation strategy was adopted, where images from DB1 were used for training and validation, while images from DB2, acquired approximately two months later, were used exclusively for testing. This evaluation protocol enabled assessment of the model's robustness under realistic biometric conditions, including variations in illumination, hand positioning, and acquisition environments.

Experimental results demonstrated that the proposed feature fusion strategy improves the discriminative capability of the identification system. The VeinFusionNet Full architecture achieved a cross-session identification accuracy of 64.11%, outperforming the baseline CNN model. The results confirm that combining spatial, frequency-domain, directional, and multi-scale edge information contributes to more reliable dorsal hand vein recognition.

Overall, the proposed system demonstrates the effectiveness of multi-feature fusion for dorsal hand vein biometrics and highlights its potential for secure and contactless human identification applications.

5.2 Future Scope

- Advanced Deep Learning Architectures

The current CNN backbone can be replaced with more advanced deep learning architectures such as ResNet, EfficientNet, DenseNet, or Vision Transformers. These architectures possess stronger feature extraction capabilities and may further improve the recognition accuracy and generalization performance of the dorsal hand vein identification system.

- Larger and More Diverse Datasets

Future work can focus on evaluating the proposed framework on larger and more diverse dorsal hand vein datasets containing greater variations in illumination, hand posture, image quality, and acquisition conditions. Such evaluations would provide a more comprehensive assessment of the robustness and scalability of the proposed system.

- Real-Time Biometric Identification

The proposed framework can be optimized for deployment in real-time biometric identification systems. By reducing computational complexity and improving inference speed, the system can be integrated into practical applications such as access control, attendance monitoring, banking security, and identity verification systems.

- Metric Learning Objectives

The current system is trained with standard Cross-Entropy Loss, which optimizes a fixed softmax boundary over 113 classes. With only six to twelve training images per identity, softmax classifiers are prone to overfitting the training session's class-specific noise rather than learning session-invariant vein embeddings. Future work should investigate metric learning objectives such as ArcFace, CosFace, or triplet loss, which directly optimize the compactness and separability of the embedding space rather than per-class decision boundaries.

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